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# AI PRACTICAL 5
# Tic Tac Toe Game
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# STEP 1 : CREATE BOARD

board = [' ' for x in range(9)]

# STEP 2 : DISPLAY BOARD

def print_board():

    print()

    print(board[0] + " | " + board[1] + " | " + board[2])

    print("--|---|--")

    print(board[3] + " | " + board[4] + " | " + board[5])

    print("--|---|--")

    print(board[6] + " | " + board[7] + " | " + board[8])

    print()

# STEP 3 : CHECK WINNER

def check_winner(player):

    win_positions = [

        [0,1,2],

        [3,4,5],

        [6,7,8],

        [0,3,6],

        [1,4,7],

        [2,5,8],

        [0,4,8],

        [2,4,6]

    ]

    for pos in win_positions:

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        if board[pos[0]] == player and \
            board[pos[1]] == player and \
            board[pos[2]] == player:

            return True

    return False

# STEP 4 : GAME LOOP

current_player = 'X'

for turn in range(9):

    print_board()

    move = int(input("Player " + current_player + " Enter Position (0-8):
"))

    if board[move] == ' ':

        board[move] = current_player

        if check_winner(current_player):

            print_board()

            print("Player", current_player, "Wins!")

            break

        if current_player == 'X':

            current_player = 'O'

        else:

            current_player = 'X'

    else:

        print("Position Already Filled")

else:

    print_board()

    print("Match Draw")

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